

AVICHAL BAWEJA

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SUMMARY

AI Engineer with strong expertise in building computer vision systems, OCR pipelines, RAG systems, and AI-powered automation solutions. Proficient in YOLOv5, OpenCV, TensorFlow, and machine learning frameworks. Experienced in developing AI agents, end-to-end ML pipelines, and deploying production-grade AI systems in fast-paced startup environments.

TECHNICAL SKILLS

AI/ML: Python, TensorFlow, PyTorch, Scikit-learn, OpenCV, YOLOv5, OCR/OMR, RAG Systems, AI Agents, LangChain

Programming: Python, C, Java, JavaScript, TypeScript, SQL, MATLAB

Frameworks & APIs: FastAPI, React, Flask, ROS, REST APIs, GraphQL, Vector Databases

Tools & DevOps: Docker, Git, Linux, Gazebo, GCP, AWS, CI/CD, Jupyter, MLflow

EXPERIENCE

Founding Engineer | *Datavedam*

Oct 2025 – Present

- Led development of AI-powered solutions including chatbot systems, automation tools, and data-driven platforms for educational institutions and government bodies.
- Built RAG pipeline that scrapes CBSE circulars, converting them into vector embeddings for intelligent querying, semantic search, and decision support.
- Built OCR and OMR pipelines for automated document scanning and exam processing with quality validation and automated workflows.
- Designed AI agent-based School ERP platform with intelligent automation for administrative tasks and automated report generation.
- Architected multi-tenant AI systems with modular ML pipeline configurations supporting customization across projects and jurisdictions.

Business Analyst | *Purshotam Profiles Pvt. Ltd.*

Jan 2025 – July 2025

- Identified new business opportunities in the solar projects sector; prepared proposals and performance reports for client engagement and decision-making.
- Collaborated with cross-functional teams to optimize sales processes and strengthen B2B client relationships.

Computer Vision Intern | *TechEagle Innovations Pvt. Ltd.*

Apr 2024 – May 2024

- Developed a CV-based drone system integrating YOLOv5 with real-time feedback control for stability, autonomous navigation, and obstacle avoidance.
- Designed an autonomous landing system in ROS/Gazebo using Python, OpenCV, and ArUco markers for precise coordinate-based landings.

IT Intern | *Sankalptaru Foundation*

Jan 2023 – Mar 2023

- Built predictive models using Python, Pandas, and scikit-learn for forecasting farm conditions; performed demand analysis with Matplotlib for agroforestry planning.

PROJECTS

Datavedam – AI Powered School ERP Platform | *Python, FastAPI, AI Agents, RAG*

- Implemented AI agents (Welfare Agent, AI Copilot) to automate admin tasks with intelligent contextual responses and workflow automation.
- Built OCR/OMR pipelines for document processing; integrated multiple APIs for end-to-end school automation and data extraction.

Drone Computer Vision System | *YOLOv5, OpenCV, ROS, Gazebo*

- Integrated YOLOv5 with real-time feedback control for drone stability, autonomous navigation, and obstacle avoidance in simulation.
- Designed autonomous landing using ArUco markers with coordinate-based precision landing achieving sub-meter accuracy.

Handwritten Text to LaTeX Converter | *Python, OCR, LLM, Image Processing*

- Built OCR + LLM pipeline to extract handwritten notes and generate accurate LaTeX documents with optimized prompts for math expressions.

EDUCATION

Christ University

B.Tech. Computer Science & Engineering, Honors in AI & ML

Bengaluru, Karnataka

2021 – 2025

Delhi Public School

Senior Secondary School (CBSE)

Agra, Uttar Pradesh

2019 – 2021